

Sasini Wanigathunga

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EDUCATION

University of Moratuwa, Sri Lanka

2021 - 2025

B.Sc. Engineering (Hons.) in Electronic and Telecommunication Engineering

GPA: 3.90/4.0

- **Dean's List:** Semesters 1, 2, 3, 4, 6, 7 and 8
- **Selected Coursework:** Deep Learning for Vision, Image Processing and Machine Vision, Pattern Recognition, Introduction to Engineering Optimization, Data Structures and Algorithms, Applied Statistics, Linear Algebra, Calculus, Neural Networks and Fuzzy Logic, Security in Cyber-Physical Systems

Sujatha Vidyalaya, Matara, Sri Lanka

2011 - 2019

GCE Advanced Level - Physical Science Stream | Island Rank: 97 | 4As

Z-score - 2.6629

- Combined Mathematics, Physics, Chemistry, General English

RELEVANT COURSEWORK

Coursera

Machine Learning Specialization (DeepLearning.AI, Stanford University)

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Retrieval Augmented Generation (RAG) (DeepLearning.AI)

Machine Learning in the Enterprise (Machine Learning on Google Cloud Specialization - Google Cloud)

EXPERIENCE

Robotic Assistance Devices, USA

Engineer I - AI Research

Sep 2025 - Present

AI Research Engineer

Mar 2025 - Sep 2025

- Contributing to the design and development of agentic AI, AI automation, and video analytics solutions for surveillance applications, with a focus on deep learning research.

Machine Learning Engineer - Intern | FcodeLabs, Sri Lanka

Nov 2023 - May 2024

- Worked on privacy-focused ML solutions, including developing a novel method for removing PII from speech data and building production-ready NLP modules for privacy data masking and LLM-based activity suggestion features.

Visiting Instructor | University of Moratuwa, Sri Lanka

Feb - Apr 2024

- Worked as a visiting instructor for EN1094: Laboratory Practice

PUBLICATIONS

Test-Time Optimization for Domain Adaptive Open Vocabulary Segmentation (*arXiv preprint, Under Review* - *Elsevier Journal of Neural Networks*)

U De Silva*, D Samaraweera*, S Wanigathunga*, K Kariyawasam*, K Ranasinghe, M Naseer, R Rodrigo

* Equal contribution

SKILLS

Programming Languages: Python, C++, C#

Software & Tools: VS Code, Git/GitHub, Hugging Face, Kaggle, Docker, AWS, MATLAB, Unity, Playwright

Frameworks: PyTorch, Tensorflow, spaCy, NLTK, Nemo, OpenCV, LangChain, LangGraph

Sports: Chess (FIDE Rating - 1219)

Languages: English (Professional proficiency), Sinhala (Native proficiency)

AWARDS

World Rank 6 in IEEE SPS Video and Image Processing (VIP) Cup 2023

Aug - Sep 2023

Team: TESSERACT | Achieved F1 score of 0.7921 for ophthalmic biomarkers detection.

IEEEExtreme 17.0 | World Rank: 389 | Country Rank: 26

Oct 2023

IEEEExtreme 16.0 | World Rank: 874 | Country Rank: 61

Oct 2022

Mahapola Higher Education (Merit) Scholarship

Aug 2021

SELECTED PROJECTS

Vision Language Models - Final Year Project

Jun 2024 - Present

Domain Adaptive Open Vocabulary Semantic Segmentation | *Computer Vision, VLMs, Prompt Learning*

- Developed a novel test-time optimization framework. Evaluated across 22 challenging domain specific datasets and obtained overall +2.03 mIoU improvement over the state-of-the-art.

Spoken Named Entity Localization | *NLP, NER, Spoken Language Understanding*

Mar 2024 - Present

- Developed a novel approach for text-independent PII removal in speech data. Achieved +4.1 frame-level F1 score, while using 67% fewer parameters and low latency over the state-of-the-art.

ProjectPulseAI 📄 | *LLMs, Groq, RAG, Next.js, Jira API, AstraDB*

Jan 2025

- Developed a LLM-powered application for project managers using Retrieval-Augmented Generation (RAG) to integrate project documents, company resources, web links, and Jira issue tracking.

Ophthalmic Biomarker Detection 📄 | *Computer Vision, Deep Learning*

Aug - Oct 2023

- Optimized the algorithm to predict the presence or absence of biomarkers on OCT scan images. Tested with different preprocessing techniques and backbones and optimized the model by changing the classification layers.

Software Design Competition 📄 | *Unity, C# Programming*

Jan - July 2024

- Designed a game using Unity with WebGL including API authentication, player profile, a questionnaire that can be opened using a web browser, dynamic and interactive game environment and a leaderboard.

Non-pipelined Single Stage (Cycle) RISC-V Processor Design 📄 | *SystemVerilog, Xilinx*

Sep - Oct 2023

- Designed a 32 bit non-pipelined RISC-V processor using Microprogramming with 3 bus structure using RV32I implementation.

Mini Weather Station 📄 | *Altium, Solidworks, C++ Programming*

July 2023

- Developed a user-friendly mini weather station with remote monitoring via web dashboard and mobile app.

Robot Design and Competition 📄 | *Arduino, Webots, C++ Programming*

Feb 2023

- Designed a simulation of a robot capable of identifying chess piece positions and performing checkmate.

LEADERSHIP EXPERIENCE

IEEE Industrial Electronics Society Student Branch Chapter - University of Moratuwa

- Secretary, Chief Editor, Assistant Editor, Member of Public Relations Team

Electronic Club - University of Moratuwa

- Chairperson for two webinars, Student Editor-in-Chief for the E-Carrier magazine, Editor, Sub Editor

AIESEC in Colombo South, Sri Lanka

- Content Specialist in Public Relations and Marketing Team, iGT International Relations and Matching Team

REFERENCES

Dr. Ranga Rodrigo

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Faculty of Engineering,

University of Moratuwa, Sri Lanka

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Dr. Sampath K. Perera

B.Sc. Eng. (Moratuwa), M.E.Sc. (Western), Ph.D. (RUB),

MIEEE

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